

Question 2

Part (a)

```
clearvars, clc
```

```
xb = [0;1;0];  
yb = [0;0;1];  
zb = cross(xb,yb)
```

```
zb = 3x1  
    1  
    0  
    0
```

```
Rab = [xb,yb,zb]
```

```
Rab = 3x3  
    0    0    1  
    1    0    0  
    0    1    0
```

```
shouldbezero = det(Rab) - 1
```

```
shouldbezero =  
    0
```

```
shouldbezero3by3 = Rab*Rab' - eye(3)
```

```
shouldbezero3by3 = 3x3  
    0    0    0  
    0    0    0  
    0    0    0
```

```
gab = [Rab,[-3.0;1.5;0.7],;0, 0, 0, 1]
```

```
gab = 4x4  
    0    0    1.0000   -3.0000  
  1.0000    0    0    1.5000  
    0    1.0000    0    0.7000  
    0    0    0    1.0000
```

```
xialt = [  
0.584897651865602 ;  
-0.589396864572260 ;  
2.047141781529606 ;  
0.449921270665848 ;  
0.584897651865602 ;  
0.674881905998771 ]
```

```
xialt = 6x1  
    0.5849  
   -0.5894  
    2.0471  
    0.4499  
    0.5849  
    0.6749
```

```
xi1 = readmatrix('xi_q2_dat.xlsx')
```

```
xi1 = 6x1
    0.5849
   -0.5894
    2.0471
    0.4499
    0.5849
    0.6749
```

```
shouldbezero6by1 = xialt - xi1
```

```
shouldbezero6by1 = 6x1
10-15 x
    0.2220
    0.1110
   -0.4441
    0.4996
    0.2220
   -0.3331
```

```
theta = pi/6;
```

$$g_{ac} = e^{\hat{\xi}_1^\theta} g_{ab} .$$

```
gtwist_num(xi1,theta)
```

```
ans = 4x4
    0.8931   -0.3022    0.3331    0.5200
    0.3727    0.9119   -0.1721   -0.3473
   -0.2518    0.2778    0.9270    0.9629
         0         0         0         1.0000
```

```
gac = gtwist_num(xi1,theta)*gab
```

```
gac = 4x4
   -0.3022    0.3331    0.8931   -2.3796
    0.9119   -0.1721    0.3727   -0.2180
    0.2778    0.9270   -0.2518    2.7839
         0         0         0         1.0000
```

Wrong:

```
wrong_a = gab*gtwist_num(xi1,theta)
```

```
wrong_a = 4x4
   -0.2518    0.2778    0.9270   -2.0371
    0.8931   -0.3022    0.3331    2.0200
    0.3727    0.9119   -0.1721    0.3527
         0         0         0         1.0000
```

Part (b)

```
Radalt = [0.135526185435788 ,    0.956172494898442 ,    -0.259551176188652 ;
-0.677630927178938 ,    0.280570323320565 ,    0.679776890017899 ;
0.722806322324201 ,    0.083752335319572 ,    0.685956679927152]
```

```
Radalt = 3x3
    0.1355    0.9562   -0.2596
   -0.6776    0.2806    0.6798
```

0.7228 0.0838 0.6860

```
Rad = readmatrix('Rad_q2_dat.xlsx')
```

```
Rad = 3x3
    0.1355    0.9562   -0.2596
   -0.6776    0.2806    0.6798
    0.7228    0.0838    0.6860
```

```
shouldbezero3by3 = Rad - Radalt
```

```
shouldbezero3by3 = 3x3
10-15 x
   -0.3331    0.3331   -0.1665
   -0.4441   -0.0555   -0.4441
    0.1110   -0.4857         0
```

```
gad = [Rad,[4.0;2.5;-1.7];0, 0, 0, 1]
```

```
gad = 4x4
    0.1355    0.9562   -0.2596    4.0000
   -0.6776    0.2806    0.6798    2.5000
    0.7228    0.0838    0.6860   -1.7000
         0         0         0    1.0000
```

MLS eq. (2.24) p. 37:

$$g_{ad} = g_{ac} g_{cd}$$

Therefore,

$$g_{cd} = g_{ac}^{-1} g_{ad} .$$

```
gcd = gac\gad
```

```
gcd = 4x4
   -0.4580   -0.0098    0.8889   -0.6952
    0.8318    0.3479    0.4325   -2.4993
   -0.3135    0.9375   -0.1512    7.8398
         0         0         0    1.0000
```

Check on answer

$$g_{ac} = g_{ab} g_{bc} .$$

Therefore,

$$g_{bc} = g_{ab}^{-1} g_{ac} .$$

```
gbc = gab\gac;
```

Then,

$$g_{ad} = g_{ab} g_{bc} g_{cd} .$$

```
dum = gab*gbc*gcd;
shouldbezero4by4 = dum - gad
```

shouldbezero4by4 = 4x4

10^{-14} x

0.0083	0.0111	-0.0056	-0.0444
0	0	0	0
-0.0111	-0.0083	0	-0.1110
0	0	0	0